

SUWALKI, Poland

WMO: 121950

Lat: 54.1308N

Lon: 22.9489E

Elev: 608

StdP: 14.38

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.5	3.5	-6.5	4.0	-1.9	-0.7	5.4	4.2	28.0	37.2	24.1	35.3	4.0	40	0.586

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	83.6	67.6	80.1	65.8	77.1	64.2	69.8	79.4	67.9	77.4	66.1	74.0	7.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	99.7	74.4	64.6	93.2	72.0	62.9	87.9	69.9	34.1	79.2	32.5	77.5	31.1	73.9	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.9	18.2	16.2	DB	-8.7	88.9	7.1	3.1	-13.8	91.1	-17.9	92.9	-21.9	94.6	-27.1	96.9	(4)
			WB	-9.1	73.0	7.0	2.2	-14.1	74.5	-18.2	75.8	-22.2	77.1	-27.2	78.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.7	24.9	26.7	33.4	45.0	54.0	59.9	64.4	63.1	54.6	44.8	36.3	28.7
	DBStd	15.91	10.56	10.11	8.13	7.78	7.33	6.05	5.60	5.48	5.90	6.94	7.51	9.73
	HDD50	3465	777	654	516	189	41	2	0	0	24	189	413	660
	HDD65	7535	1242	1074	980	602	348	177	81	103	315	626	862	1125
	CDD50	1543	0	0	1	38	165	297	448	405	161	28	0	0
	CDD65	140	0	0	0	0	7	23	64	44	2	0	0	0
	CDH74	1433	0	0	0	16	95	255	585	445	37	0	0	0
	CDH80	312	0	0	0	1	8	46	152	101	4	0	0	0
Wind	WSAvg	7.8	8.8	8.6	8.8	7.8	7.4	7.3	6.8	6.3	6.6	7.7	8.4	8.7
Precipitation	PrecAvg	23.4	1.3	1.0	1.3	1.5	2.1	2.9	3.1	2.7	2.0	1.9	2.0	1.6
	PrecMax	32.6	4.1	2.1	3.1	3.7	4.1	5.9	6.9	7.1	4.9	7.6	4.8	3.1
	PrecMin	15.5	0.3	0.0	0.2	0.2	0.6	0.9	0.7	0.4	0.5	0.1	0.9	0.3
	PrecStd	3.7	0.7	0.5	0.7	0.7	0.9	1.2	1.7	1.6	1.0	1.4	0.9	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.3	46.2	58.9	75.8	81.0	85.3	88.8	88.0	78.7	66.3	53.4	47.7
		MCWB	43.6	43.3	48.4	59.6	64.4	68.6	69.4	69.7	63.4	57.5	50.5	45.9
	2%	DB	41.6	43.4	53.1	69.3	76.6	80.3	84.3	82.5	73.0	61.5	50.2	44.6
		MCWB	40.2	41.0	45.2	55.4	62.4	65.7	68.2	66.8	61.9	55.8	48.2	43.0
	5%	DB	39.1	40.5	48.4	64.1	72.7	76.6	80.3	78.9	69.0	57.8	47.7	42.1
		MCWB	37.8	38.7	42.8	51.7	60.0	63.6	67.2	65.1	59.8	53.8	46.0	40.8
	10%	DB	36.6	38.2	44.3	59.5	68.8	73.1	76.7	75.4	65.5	55.0	45.8	39.5
		MCWB	35.6	36.7	40.7	49.6	57.9	61.8	65.2	63.7	57.9	52.2	44.6	38.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.1	43.9	49.8	60.6	66.5	70.9	72.8	71.7	66.1	59.4	51.4	46.3
		MCDB	45.2	45.5	56.1	74.5	77.6	81.4	83.4	82.4	73.4	64.3	53.0	47.3
	2%	WB	40.5	41.3	46.2	56.5	63.8	67.8	70.2	69.1	63.1	56.7	48.5	43.4
		MCDB	41.3	42.9	51.4	66.4	73.1	77.3	79.5	78.9	70.4	59.8	49.8	44.4
	5%	WB	37.8	38.9	43.4	53.4	61.5	65.3	68.3	67.0	60.9	54.4	46.4	40.9
		MCDB	38.7	40.4	47.4	61.8	70.1	73.5	77.7	75.7	67.0	57.2	47.5	42.0
	10%	WB	35.7	36.8	40.9	50.7	59.2	63.1	66.6	65.1	58.9	52.3	44.7	38.5
		MCDB	36.5	38.0	44.1	57.7	67.2	70.7	74.5	72.7	64.3	54.9	45.7	39.5
Mean Daily Temperature Range	5% DB	MDBR	8.0	9.4	13.0	18.4	20.0	18.9	18.4	19.2	16.7	12.4	7.4	7.4
		MCDBR	7.3	9.3	19.4	26.6	25.4	24.6	24.2	24.9	22.6	17.2	9.3	7.4
		MCWBR	6.9	7.7	12.8	14.4	13.2	12.2	10.9	11.5	12.7	11.5	7.9	6.9
	5% WB	MCDBR	7.0	8.8	16.9	24.0	22.7	21.5	21.0	21.0	19.9	15.0	8.7	7.4
		MCWBR	6.7	7.7	12.0	14.0	12.8	12.0	10.4	10.7	11.9	10.8	7.7	7.1
Clear-Sky Solar Irradiance	taub	0.298	0.329	0.365	0.408	0.397	0.390	0.426	0.420	0.380	0.356	0.339	0.297	
	taud	2.207	2.194	2.218	2.193	2.288	2.333	2.250	2.264	2.347	2.399	2.291	2.244	
	Ebn at Noon	203	236	254	259	269	272	259	252	248	227	186	178	
	Edh at Noon	20	28	35	40	39	37	40	37	30	23	19	16	
All-Sky Solar Radiation	RadAvg	195	430	825	1295	1683	1754	1630	1384	973	492	183	124	
	RadStd	36	76	122	168	172	147	167	133	123	94	20	18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+117	N/A

Nomenclature: See separate page